

ANTHELMINTIC AGENTS

- Anthelmintics are drugs that have the ability to get rid of the parasitic worms or helmenths from the human or other animal bodies.
- Prevention of human helminthic infection all over the world represents the major WORLD HEALTH ORGANIZATION (WHO) problem.

- The anthelmintic agents are drugs which could get rid of the following kinds of worms:
 - ✓ Nematodes (Pin worm, Filarial worm, Hook worm, Round worm).
 - ✓ Cestodes (Fish tapeworm, Beef tapeworm, Pork tapeworm).
 - ✓ Trematodes (Schistosoma haematobium, Schistosoma mansoni, & Schistosoma japonicum).

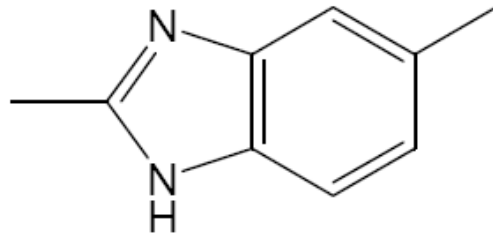
DRUGS ACTING ON NEMATODES:

- a) Benzimidazoles**
- b) Imidazothiazole.**
- c) Piperazines.**
- d) Vinyl-pyrimidines (Pyrentel)**

Anti-NEMATODES:

A) BENZIMIDAZOLES

Benzimidazoles are broad- spectrum group of drugs with high activity against GIT helminthes.



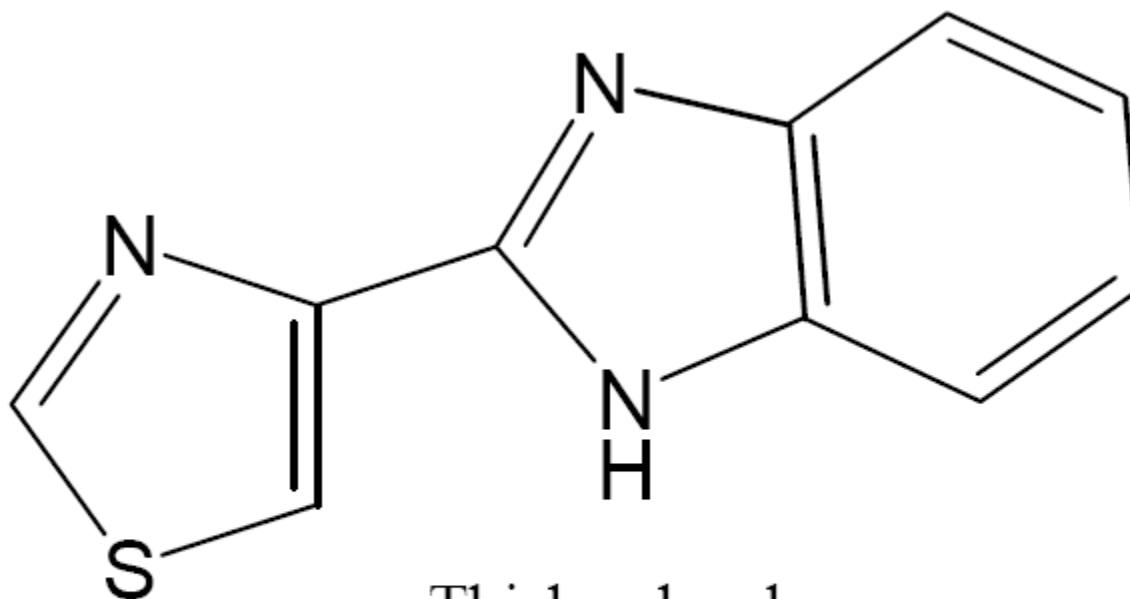
Benz-imidazoles

Their mode of action

- inhibiting the production of microtubules via binding to colchicine binding-site of β -tubulin and thereby blocking polymerization of tubulin dimers in the intestinal cells of parasites.
- Most of the benzimidazoles are weakly soluble in the intestinal fluids to exert a useful local action in the GIT infections.

Examples of benzimidazoles;

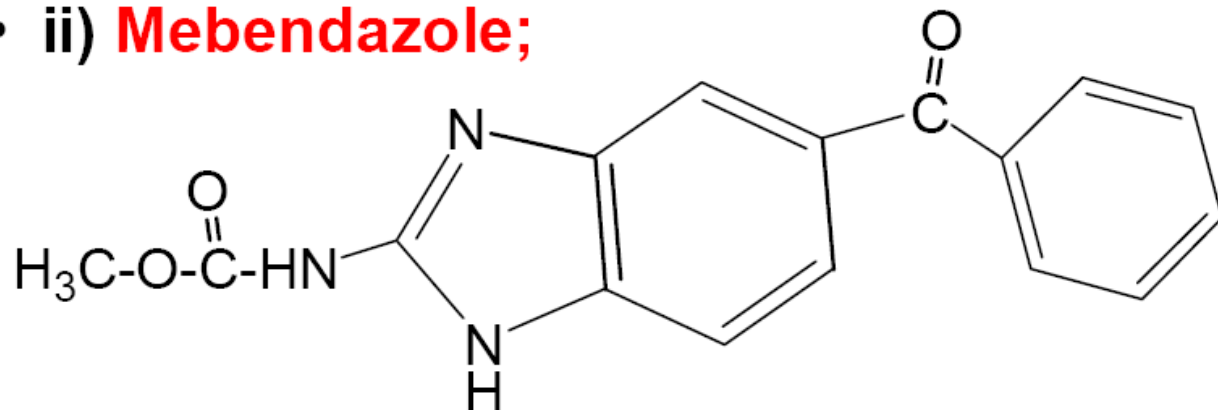
i) Thiabendazole (Mintezole)



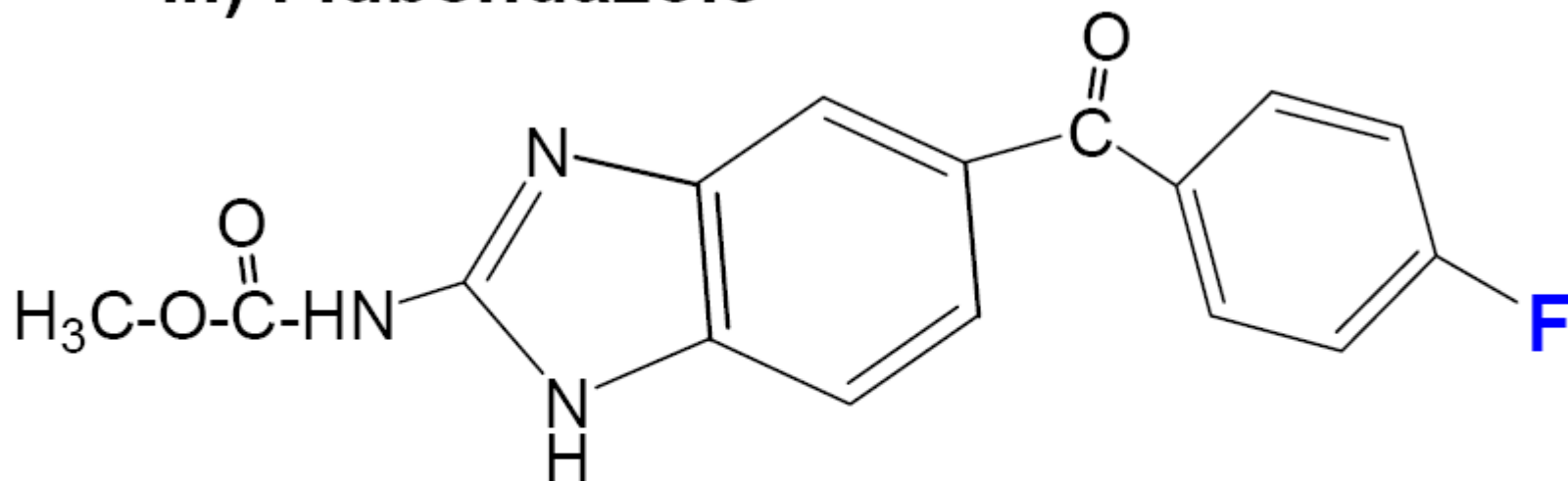
Thiabendazole

2-(4-Thiazolyl)benzimidazole

- ii) **Mebendazole;**

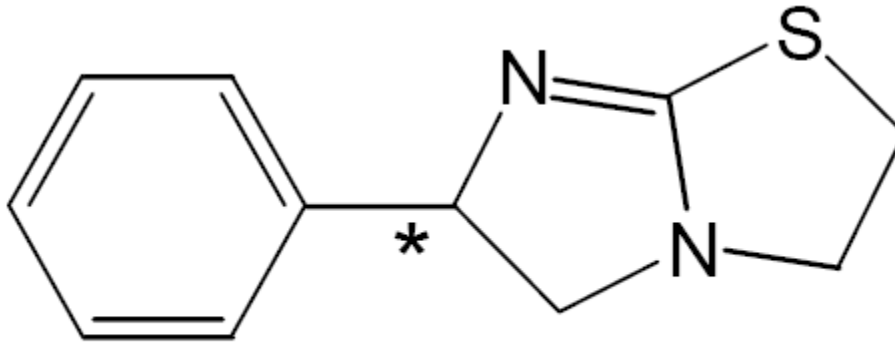


iii) Flubendazole



b) IMIDAZOTHIAZOLE

- **Levamisole (KETRAX)**

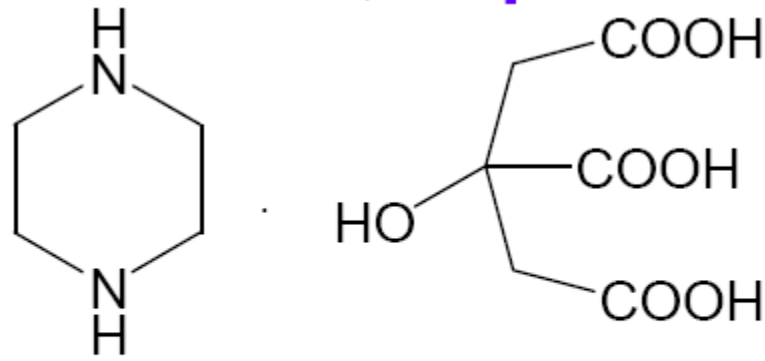


Levamisole

**The active form is the Levo isomer named:
(-),2,3,5,6-Tetrahydro-6-phenylimidazo-(2,1-
b)thiazole**

**This drug is the drug of choice to treat Ascariasis,
and ancylostomiasis.**

c) PIPERAZINES,



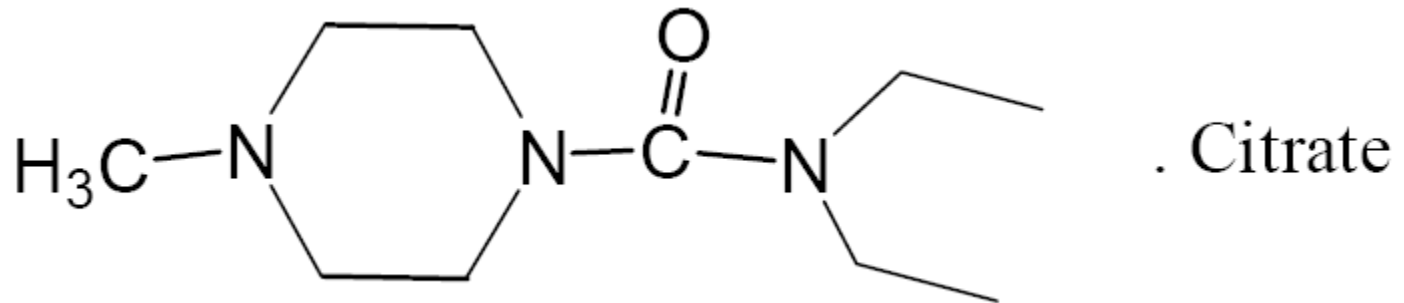
Piperazine Citrate

They are the drug of choice to get rid of pinworms and roundworms and various nematodes.

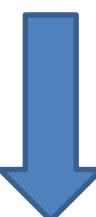
The mode of action;

Blocking of the neuromuscular transmission in the worms causing a flaccid paralysis of it ,which dislodges it from GIT and expelled in the face.

ii) Diethyl Carbamazine Citrate (DEC)
(Hetrazan)

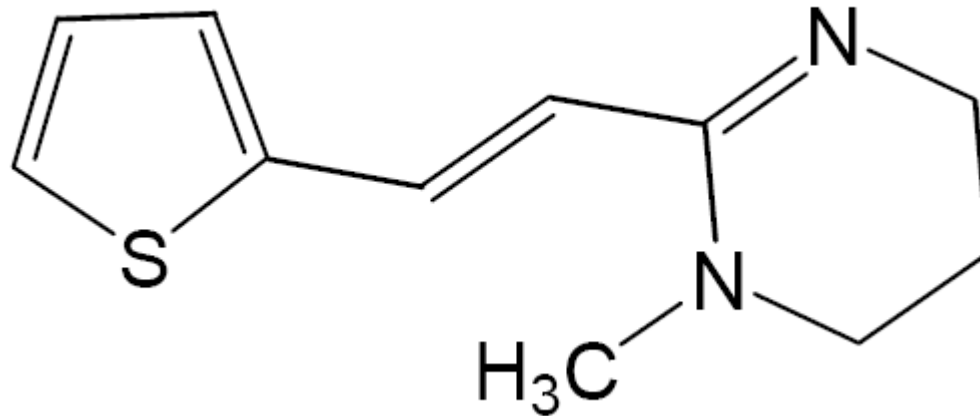


DEC is active against FALARIASIS and MICROFALARIASIS.

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- The mode of action of DEC is suggested to have one of the following:

- 1) Complexation with the filarial antigen which inhibit blood platelets.
- 2) Binding to the helminthes proteins; tubuline and preventing its polymerization to microtubuline.
- 3) Interfere with arachidonic acid metabolism. Hence this drug has antiinflammatory activity by inhibiting COX-II enzyme.

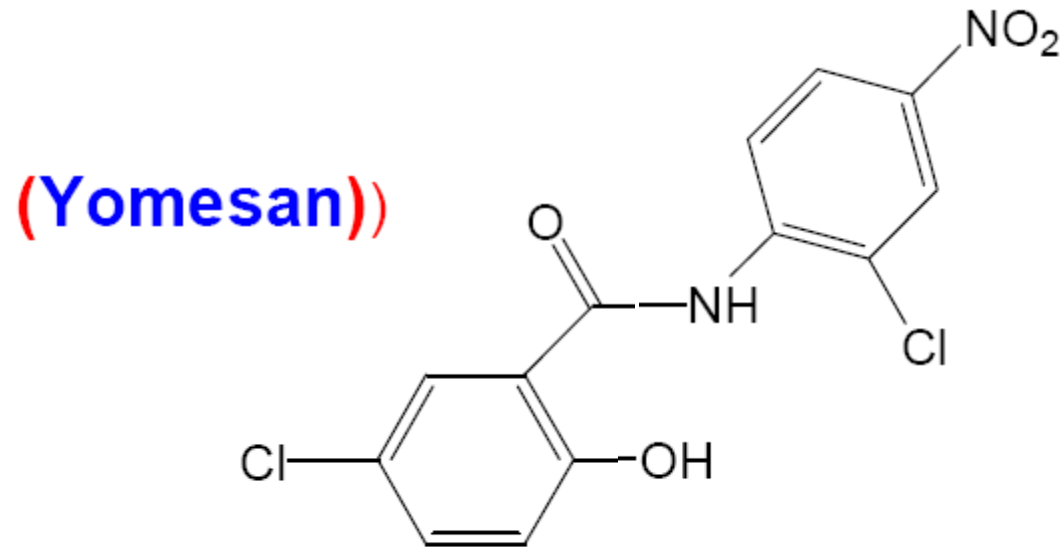
d) PYRANTEL PAMOATE:



- It is a depolarizing neuromuscular blocking agent that causes spastic paralysis .
- It is used in the treatment of infestation caused by pinworms and Ascariasis.
- Because its action opposes that of piperazine, the two helmentics should not be used together.

DRUGS ACTING ON CESTODES

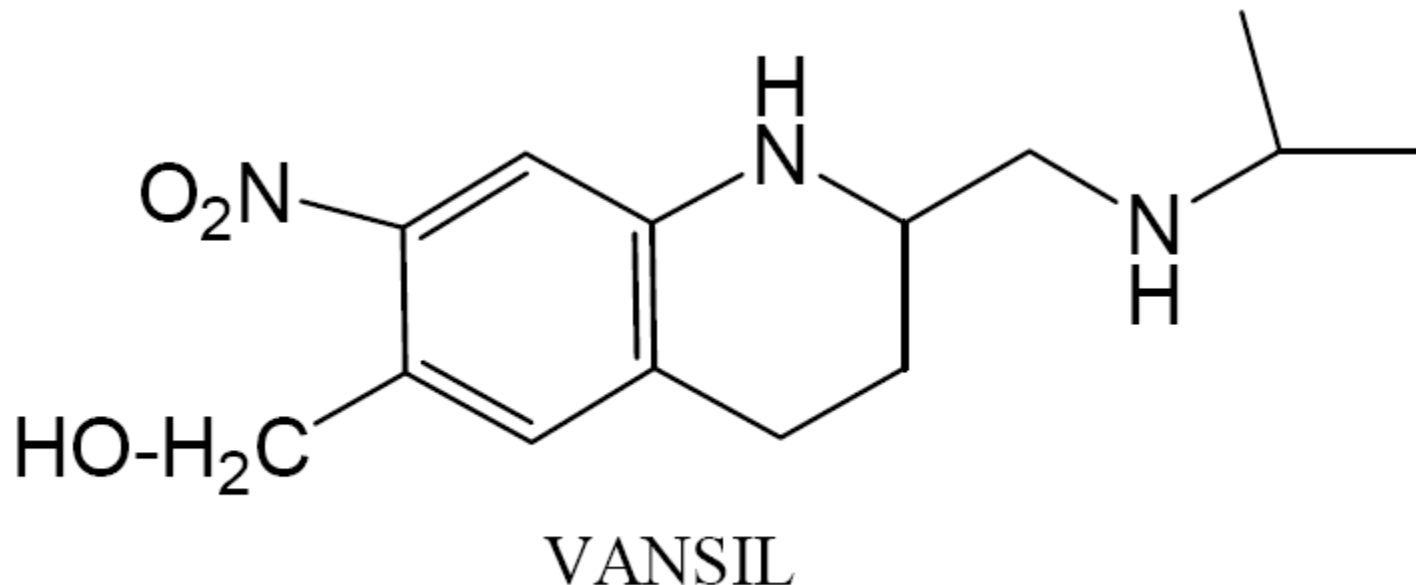
- A. Benzimidazoles (as mentioned before)
- B. Miscellaneous (the most important is
Niclosamide (Yomesan))



Drugs acting on TREMATODES

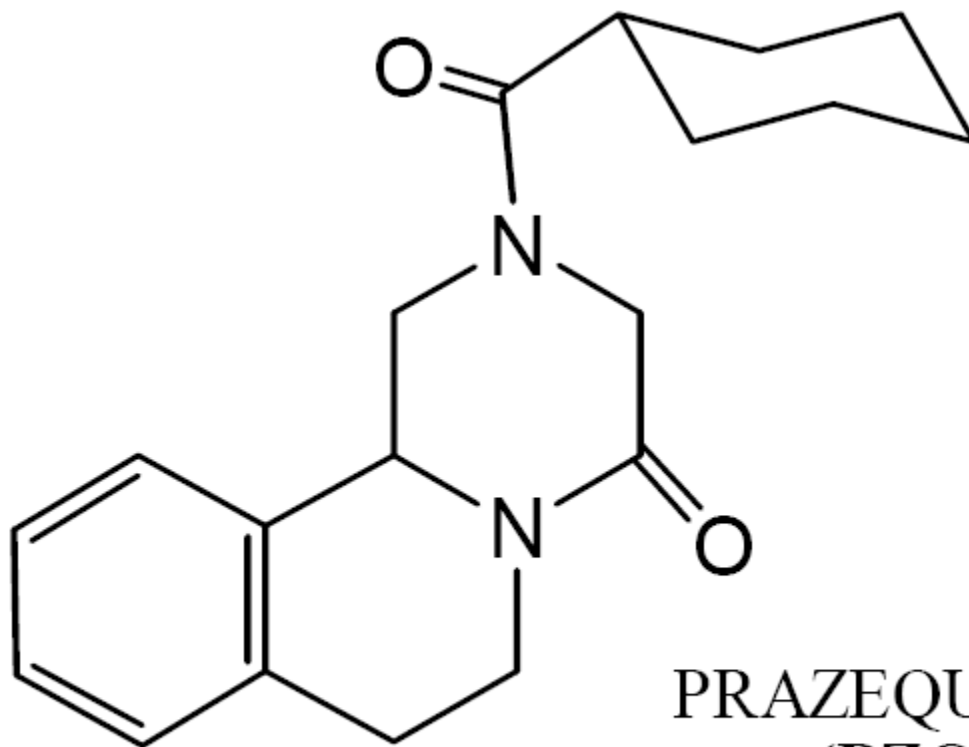
(Antischistosomal or Antibilharzial agents)

A) Oxamniquine (VANSIL)



- 1,2,3, 4 -Tetrahydro-2-(isopropyl- amino)methyl
-7 – nitro -6-quinolinemethanol (Vansil)
- It inhibit nucleic acid synthesis of Schistosomes
- The 6- hydroxymethyl group is critical for activity
 - The metabolic activation of its precursor 6- methyl group is vital
- Metabolized in to 6 carboxylic derivative and
inactive
- Has good oral bioavailability

b) PRAZEQUANTEL (Biltricide , PZQ)



PRAZEQUANTIL
(PZQ)

The Levo-isomer of PZQ is the active enantiomers.

Mode of PZQ

- Increases cell membrane permeability, resulting in loss of extracellular calcium. Muscle contraction and ultimate paralysis and hence their ease eradication.
- PZQ is the drug of choice for treating Schistosomiasis and liver Flukes (Trematodes and Cestode infections).